Equatorial Waves

Self-Paced Class Activities

Learning goals. By the end of class you should be able to (i) explain why tropical weather variability is often more apparent in convection than in large-scale temperature or pressure fields; (ii) describe how OLR is used to identify organized tropical convection; (iii) recognize key equatorial wave families from their spatial structure; and (iv) discuss why equatorial waves matter for tropical cyclones and weather prediction.

Activity 1 — What makes tropical weather different? (5 min)

Refer to lecture slides 1–2 and the global forecast fields discussed in class.

1. **What contrasts do you observe between day-to-day tropical and extratropical variability in pressure, temperature, wind, and precipitation?**

2. **If large-scale pressure and temperature gradients are weak in the tropics, does that imply weak weather variability? Explain.**

Key idea to record: Tropical variability is especially apparent in _____, even when broad temperature and pressure patterns change relatively slowly.

Activity 2 — Reading the OLR animation (8 min)

Refer to lecture slides 3–4. Observe the OLR animation shown by the instructor.

1. **Why do cold, high cloud tops generally correspond to low outgoing longwave radiation (OLR)? What does low OLR suggest about convection?**

2. **Describe the organization and movement of convective regions. Do all convective features travel in the same direction?**

3. **What changes when a 40-day mean is removed from the OLR field? What is easier to see, and what information could be lost?**

Activity 3 — From messy convection to organized waves (8 min)

Refer to lecture slides 5–8 and compare the original and filtered wave-mode animations.

1. **Why might the unfiltered tropical convection field appear disorganized even if coherent waves are present?**

2. **After separating wave modes, what repeated patterns do you notice in the location, spacing, and movement of active and suppressed convection?**

3. **The lecture compares active/suppressed tropical convection loosely to extratropical troughs/ridges. Where does this analogy help, and where might it be misleading?**

Lecture notes: Four broad wave families identified in this survey are:

1. _____
2. _____
3. _____
4. _____

Activity 4 — Equatorial Rossby waves (8 min)

Refer to the theoretical Rossby-wave diagram (lecture Fig. 1), the observed 850-hPa wind analysis (Fig. 2), and twin-tropical-cyclone image (Fig. 3).

1. **In Fig. 1, identify the paired circulation structures on either side of the equator. How does the diagram suggest that the disturbance is equatorially trapped?**

2. **Compare the idealized pattern in Fig. 1 with the observed twin-vortex wind structure in Fig. 2. What is similar? What is less orderly in reality?**

3. **How does the twin-cyclone example (Fig. 3) illustrate cross-equatorial organization? Does the image alone prove that an equatorial Rossby wave caused the cyclones?**

Activity 5 — Comparing wave families (10 min)

Refer to the theoretical patterns in lecture Figs. 1, 4, 5, and 7, and the observed mixed Rossby–gravity-wave image (Fig. 6). The diagrams show geopotential contours and regions of convergence/divergence.

1. **Equatorial Rossby versus mixed Rossby–gravity waves: How does the north–south symmetry of the circulation differ in the idealized diagrams?**

2. **Look at the inertio-gravity wave panels (Fig. 4). Compared with Rossby waves, what aspects of the circulation and convergence/divergence seem especially prominent?**

3. **Look at the Kelvin-wave diagram (Fig. 7). What is distinctive about the near-equatorial confinement and the wind pattern?**

4. **In the observed mixed Rossby–gravity wave (Fig. 6), what does the satellite image suggest about the relationship between convective clouds and wave organization?**

Quick comparison notes:

Wave family	**Distinguishing visual or physical characteristic**
Equatorial Rossby	
Inertio-gravity	
Mixed Rossby–gravity	
Kelvin	

Activity 6 — Why should we care? (6 min)

Refer to lecture slide 18, which links equatorial waves to tropical cyclones and broader atmospheric prediction.

1. **Suggest two ways that a propagating wave might make the large-scale environment more or less favorable for tropical cyclone formation. Think about winds, moisture, and convection.**

2. **Why could errors in simulating tropical wave activity affect extratropical forecasts? Explain a plausible chain of connections.**

Synthesis — An interpretive challenge (5 min)

A satellite animation shows alternating regions of enhanced and suppressed convection moving along the equator, while the large-scale temperature field changes little.

1. **Why is this behavior not contradictory?**

2. **What additional observations would you request before identifying the underlying wave mode?**

Reflection (2 min)

One idea from today's lecture that is now clear:

One idea that remains unclear or needs further explanation: